3.12 Low-power direct memory access controller (LPDMA)

The low-power direct memory access (LPDMA) controller is a bus master and system peripheral. The LPDMA is used to perform programmable data transfers between memory-mapped peripherals and/or memories via linked-lists, under the control of an off-loaded CPU.

The LPDMA main features are:

- Single bidirectional AHB master
- Memory-mapped data transfers from a source to a destination:
  - Peripheral to memory
  - Memory to peripheral
  - Memory to memory
  - Peripheral to peripheral
- Transfers arbitration based on a 4-grade programmed priority at the channel level:
  - One high-priority traffic class for time-sensitive channels (queue 3)
  - Three low-priority traffic classes with a weighted round-robin allocation for non-time-sensitive channels (queues 0, 1, 2)
- Per channel event generation on any of the following events: transfer complete, half transfer complete, data transfer error, user setting error, link transfer error, completed suspension, and trigger overrun
- 16 concurrent LPDMA channels (8-channel LPDMA1, 8-channel LPDMA2):
  - Intrachannel LPDMA transfers chaining via programmable linked-list into memory, supporting two execution modes: run-to-completion and link step mode
  - Intrachannel and interchannel LPDMA transfers chaining via programmable LPDMA input triggers connection to LPDMA task completion events
- Per linked-list item within a channel:
  - Separately programmed source and destination transfers
  - Programmable data handling between source and destination: byte-based padding or truncation, sign extension, and left/right realignment
  - Programmable number of data bytes to be transferred from the source, defining the block level
  - Linear source and destination addressing: either fixed or contiguously incremented addressing, programmed at a block level, between successive single transfers
  - Programmable LPDMA request and trigger selection
  - Programmable LPDMA half-transfer and transfer-complete events generation
  - Pointer to the next linked-list item and its data structure in memory, with automatic update of the LPDMA linked-list control registers
- Debug:
  - Channel suspend and resume support
  - Channel status reporting and event flags
- Privileged/unprivileged support:
  - Support for privileged and unprivileged LPDMA transfers, independently at the channel level
  - Privileged-aware AHB slave port

Table 2. LPDMA1 and LPDMA2 channels implementation and usage

Channel x	Hardware parameters		Features
	dma_fifo_size[x]	dma_addressing[x]	
x = 0 to 7	0	0	Channel x (x = 0 to 7) is implemented with: • no FIFO. Only a single source transfer cell is internally registered. • fixed/contiguously incremented addressing